According to ***"DI WU,HAO WANG,RAZAK SEIDU" in [3]*** Drinking Water Supply (DWS) is one of the most critical and sensitive systems to maintain city operations globally. In Europe, the contradiction between the fast growth of population and obsolete urban water supply infrastructure is even more prominent. The high standard water quality requirement not only provides convenience for people's daily life but also challenges the risk response time in the systems. Prevalent water quality regulations are relying on periodic parameter tests. This brings the danger in bacteria broadcast within the testing process which can last for 24-48 hours. In order to cope with these problems, we propose a tensor model for water quality assessment. This model consists of three dimensions, including water quality parameters, locations and time. Furthermore, we applied this model to predict water quality changes in the DWS system using a Random Forest algorithm. For a case study, we select an industrial water supply system in Oslo, Norway. The preliminary results show that this model can provide early warning for water quality risks.

According to ***"LIJU PHILIP,RAKESH VS,SREESH,HEMALATHA" in [4]*** Places where the water resources are scarce, there is a requirement for water conservation. It is seen that water is wasted due to unknown leaks and mismanagement and this problem needs to be addressed. This happens in houses, public/private organizations, educational institutions etc. The design and implementation of an embedded system is proposed to address this problem. It gives an optimized solution which reduces water wastage to a great extent thereby aiding water conservation. A day is split into different time intervals during which different water flow rates for each interval can be easily set by the user. Whenever the water flow rate exceeds the set limit, warnings are generated by the system. An additional advantage of the proposed system is that it ensures electrical energy conservation.

According to ***"ZHONG MA,SIYANG LUI,MIAOMIAO WANG" in [5]*** The System of Environmental and Economic Accounting for Water (SEEAW) is an accounting framework exclusively on water resources, which was published by United Nations Statistics Division. The ability to address jointly the environmental, economic, and social